

BUSINESS UNDERSTANDING REPORT

Customer Churn Analysis

Prepared by: Hudu Yusuf Ibrahim

Project: Customer Churn Analysis

Dataset: Telco Customer Churn Dataset

Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

1. Introduction

Customer retention is a critical factor for the success of subscription-based businesses such as telecommunications companies. While acquiring new customers supports business growth, retaining existing customers is generally more cost-effective and contributes to stable long-term revenue. Understanding why customers leave enables organisations to improve customer satisfaction, strengthen loyalty, and make informed business decisions.

This project analyses the **Telco Customer Churn** dataset to identify the factors associated with customer churn and provide data-driven recommendations that support customer retention.

2. Business Problem

Customer churn is a major business challenge because it reduces recurring revenue, increases customer acquisition costs, and can negatively affect profitability. Although the company records whether customers leave, it needs to understand the characteristics and behaviours associated with churn in order to develop effective retention strategies. This project aims to analyse customer data to identify high-risk customer segments and provide actionable insights that can help management improve customer retention.

3. Stakeholders

The findings from this project will benefit several stakeholders within the organisation:

- **Executive Management:** Uses the findings to support strategic decision-making and improve business performance.
- **Marketing Team:** Designs targeted customer retention campaigns for high-risk customer segments.
- **Customer Service Team:** Identifies customers who may require additional support to improve satisfaction and retention.
- **Business Analysts:** Uses the analysis to monitor customer behaviour and support future business decisions.

4. Project Objectives

The objectives of this project are to:

- Understand the characteristics of the customer base.
- Identify customer segments with higher churn.
- Examine whether contract type influences customer retention.
- Analyse the relationship between customer tenure and loyalty.
- Investigate how internet services, value-added services, and payment methods relate to customer churn.

- Develop evidence-based recommendations that support customer retention.

5. Business Questions

The analysis seeks to answer the following business questions:

1. What does the customer base look like?
2. Which segments have the highest churn?
3. Does contract type influence retention?
4. Does tenure affect loyalty?
5. Which services influence churn?
6. Which payment methods have higher churn?
7. What actions should management take?

6. Success Criteria

The project will be considered successful if it:

- Identifies customer segments associated with higher churn.
- Explains the relationship between customer characteristics and retention.
- Produces clear, evidence-based insights from the data.
- Provides practical recommendations that management can use to improve customer retention and support business decision-making.

7. Key Insights

The completed analysis identified several important patterns that informed the business recommendations presented in this project:

- Overall churn stands at 26.5%, meaning over a quarter of customers have left the company.
- Contract type is strongly linked to retention: month-to-month customers churn at 42.7%, versus 11.3% (one-year) and 2.8% (two-year).
- Churn is concentrated among newer customers (highest in the first 12 months) and among those with higher monthly charges and lower use of add-on services such as Online Security and Tech Support.

8. Expected Business Value

The project is expected to help the organisation understand customer behaviour, identify high-risk groups, and support evidence-based decision-making that reduce churn and strengthen long-term business performance.

9. Conclusion

This Business Understanding Report defines the business problem, project objectives, stakeholders, success criteria, and key business questions for the Customer Churn Analysis project. It provides the foundation for the subsequent stages of dataset inspection, exploratory data analysis, and the development of business insights and recommendations aimed at improving customer retention.